

20MCA245 MINI PROJECT

SYNOPSIS

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Title: SwapSphere – A Smart Barter Platform
with Graph-Based Exchange Matching

Synopsis Approved by Guide: Yes / No

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Remarks if any:

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SYNOPSIS

SwapSphere – A Smart Barter Platform with Graph-Based Exchange Matching

1. Problem Statement

Most existing online marketplaces focus on buying and selling rather than direct item exchange, making barter inconvenient and inefficient. Users often face difficulties in finding suitable exchange partners while also dealing with trust and transparency issues during exchanges. SwapSphere aims to overcome these challenges by providing a secure platform that identifies suitable direct and three-way exchange opportunities using graph-based matching techniques without the use of monetary transactions.

2. Objectives

- To develop a secure web-based platform that enables users to exchange items directly without monetary transactions.
- To implement a graph-based exchange matching engine that supports both direct (two-way) and three-way exchanges, improving the chances of successful item exchanges.
- To build a transparent and trustworthy barter ecosystem by incorporating a reputation system, user ratings and reviews, and an Exchange Room for communication and coordination between exchange participants.

3. Proposed System

SwapSphere is a full-stack web application that modernizes the traditional barter system by enabling users to exchange items directly instead of buying and selling them. Users can register using either Google Sign-In or email authentication, create personal profiles, and list items they wish to exchange. While creating a listing,

users can specify multiple acceptable items they are willing to receive in return, arranged according to their preference.

To improve transparency and trust, users can upload clear images and videos of their items along with descriptions. The platform also maintains a reputation system based on completed exchanges, user ratings, reviews, and exchange history, helping users make informed decisions before initiating an exchange.

The core component of SwapSphere is a graph-based Exchange Matching Engine that models exchange requests as a graph and identifies valid exchange opportunities using cycle detection. The system supports both direct exchanges between two users and automated three-way exchange cycles, increasing the chances of successful exchanges when a direct match is unavailable. Once all participants accept an exchange proposal, the platform creates an Exchange Room where users can communicate, coordinate meeting details, and confirm the successful completion of the exchange. The proposed system aims to provide a secure, efficient, and sustainable approach to modern barter while encouraging the reuse of existing resources.

4. Required Technologies

a. Database / AI / ML

- Database: MongoDB will be used to store user information, item listings, exchange requests, reputation data, ratings, reviews, and exchange history.
- AI / ML: Not applicable. The system uses graph-based algorithms and cycle detection for exchange matching instead of Artificial Intelligence or Machine Learning.

b. Frontend

- React.js
- HTML5
- CSS3
- Bootstrap
- JavaScript

The frontend will provide a responsive and intuitive interface for user authentication, item listing, searching, exchange management, and Exchange Rooms.

c. Backend

- Node.js

- Express.js
- RESTful APIs
- JWT Authentication
- Email & Password Authentication
- Google OAuth Authentication

The backend will manage authentication, business logic, graph-based matching, reputation management, exchange workflows, and communication with the database.

d. Other Tools

- Git & GitHub (Version Control)
- Visual Studio Code
- Postman (API Testing)
- MongoDB Atlas

5. Expected Outcome

The proposed system is expected to provide a secure and user-friendly platform that simplifies direct item exchange without monetary transactions. The graph-based matching engine and reputation system will improve the efficiency of finding suitable exchange partners while increasing user confidence during exchanges. By supporting both direct and three-way exchanges, SwapSphere aims to promote sustainable resource utilization and offer a practical digital solution for modern barter.

6. Tentative Work Plan

Timeline	Activities
06/07/2026 – 13/07/2026	Requirement analysis, literature survey, project planning, and UI wireframe design.
14/07/2026 – 31/07/2026	Database design, authentication module, user profile management, and item listing functionality.
01/08/2026 – 08/08/2026	Search functionality, filtering, reputation system, and preparation for the first evaluation (50% completion).
09/08/2026 – 15/09/2026	Development of the graph-based exchange matching engine, direct exchange workflow, and three-way exchange support.
16/09/2026 – 05/10/2026	Implementation of the Exchange Room, testing, debugging, and feature refinement.
06/10/2026 – 16/10/2026	Documentation, deployment, final testing, report preparation, and final project evaluation.